Constructor University Bremen

Natural Science Laboratory
Electronics

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Lab Experiment 3 – Fourier Series and Fourier Transform

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Place of execution	: Teaching Lab 53
Date of execution	: November 5, 2025

1. Introduction

The purpose of this experiment was to study and analyze the fundamental principles of amplitude modulation and to gain a deeper understanding of how information can be transmitted through variations in the amplitude of a carrier signal. Both theoretical and practical aspects of amplitude modulation were explored during the experiment to examine how different parameters affect the behavior of the modulated wave.

The experiment was divided into three main parts. In the first part, amplitude-modulated signals were analyzed in the time domain to observe the relationship between the modulation index and the shape of the signal envelope. The second part focused on the frequency domain analysis, where the Fast Fourier Transform function of the oscilloscope was used to identify the carrier and sideband components of the modulated signal. In the final part, the demodulation process was carried out to recover the original message signal using an envelope detector and a multi-stage low-pass filter.

Throughout the experiment, different types of amplitude modulation such as double sideband, double sideband suppressed carrier, and single sideband modulation were examined, and their frequency characteristics were studied. Finally, the experimental measurements were compared with the theoretical calculations to evaluate the accuracy of the results and to better understand the performance and limitations of amplitude modulation systems.

1.1. Prelab – Problem 1 – Single Frequency Amplitude Modulation

a) Derive an expression describing the modulation index m as a function of the modulation envelope, (use Amin and Amax!).
Modulating Signal: $x(t) = A_m \cos(2\pi f_m t)$
Carrier Signal: $y(t) = A_C(1 + m\cos(2\pi f_m t))\cos(2\pi f_C t)$
Max Amplitude: $A_{max} = A_C(1 + m)$
Min Amplitude: $A_{min} = A_C(1 - m)$
From here we can derive:

$$A_C = \frac{A_{max}}{1+m} \quad and \quad A_C = \frac{A_{min}}{1-m}$$

We can find m as:

$$m = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

b) Derive an expression describing the ratio of the total sideband power to the total power $r_P = P_S/P_{tot}$ in the modulated wave delivered to a load resistor. Express the ratio in terms of the modulation index.
As we know:

$$P_T = P_S + P_C \quad and \quad P_T = P_C(1 + \frac{m^2}{2})$$

Therefore:

$$P_S = \frac{P_C m^2}{2}$$

So,

$$r_P = \frac{P_S}{P_T} = \frac{\frac{P_C m^2}{2}}{P_C(1 + \frac{m^2}{2})} = \frac{m^2}{2 + m^2}$$

c) Calculate the ratio r_P assuming a modulation index of 100%.

$$m = 100\% = 1$$

$$r_P = \frac{m^2}{2 + m^2} = \frac{1}{2 + 1} = \frac{1}{3}$$

d) A Carrier $V_C(t) = 5\cos(20000\pi t)$ is AM modulated by as signal $V_m(t) = 2 + \cos(2000\pi t)$ Calculate the ratio r_P. How would you change the input signals to maximize the side band to total power ratio?

Using:

$$y(t) = A_C(1 + kA_m\cos(2\pi f_m t))\cos(2\pi f_C t)$$

Plugging Values:

$$y(t) = A_C(1 + k(2 + A_m\cos(2\pi f_m t)))\cos(2\pi f_C t)$$

Since $kA_m = m$

$$y(t) = A_C(1 + 2 + m\cos(2\pi f_m t)))\cos(2\pi f_C t)$$

Using trigonometric identity:

$$y(t) = (A_C + 2)\cos(2\pi f_C t) + \frac{mA_C}{2}\cos(2\pi f_m t + 2\pi f_C t) + \frac{mA_C}{2}\cos(2\pi f_m t - 2\pi f_C t)$$

The total Power:

$$P_{tot} = P_C + P_U + P_L$$

$$P_{tot} = \frac{V_C^2}{R} + \frac{V_U^2}{R} + \frac{V_L^2}{R} = \frac{(A_C + 2)^2}{2R} + \frac{m^2 A_C^2}{8R} + \frac{m^2 A_C^2}{8R}$$

The final value for r_P was calculated as:

$$r_P = \frac{P_S}{P_{tot}} = \frac{\frac{m^2 A_C^2}{8R} + \frac{m^2 A_C^2}{8R}}{\frac{(A_C + 2)^2}{2R} + \frac{m^2 A_C^2}{8R} + \frac{m^2 A_C^2}{8R}} = \frac{\frac{m^2 A_C^2}{4R}}{\frac{(A_C + 2)^2}{2R} + \frac{m^2 A_C^2}{4R}} = \frac{A_C^2 m^2}{2(A_C + 2)^2 + A_C^2 m^2}$$

After Plugging the values:

$$r_P = \frac{1}{2(5 + 2)^2 + 1} = \frac{1}{99} = 0.01$$

1.2. Prelab – Problem 2 – Amplitude Demodulation

Use MatLab to generate an AM modulated signal with a carrier frequency f = 20KHz and am = 5V. Modulate with a sine wave f = 500Hz. Use a modulation index m = 50%. Simulate the demodulation of this AM signal with Matlab. Show the following steps:

a) Plot the modulated signal in time and frequency domain.

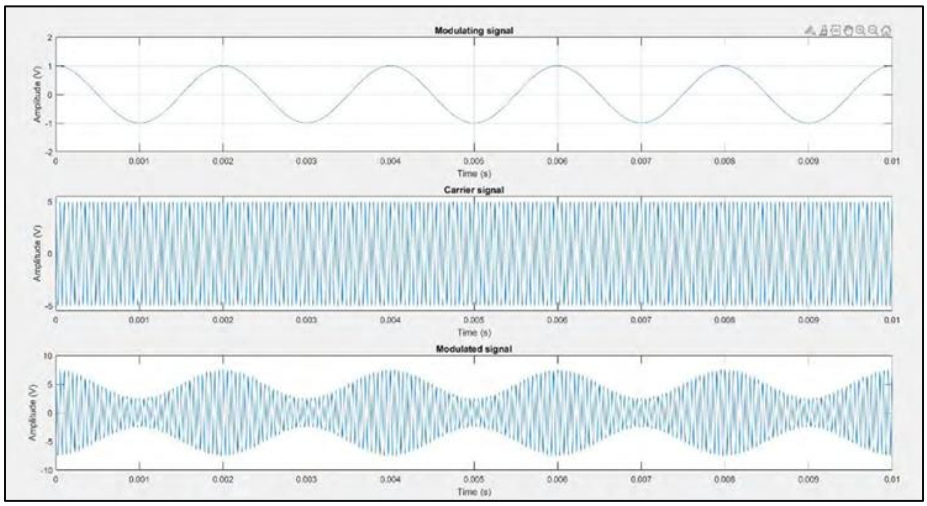

Figure 1. Modulated Signal in time domain

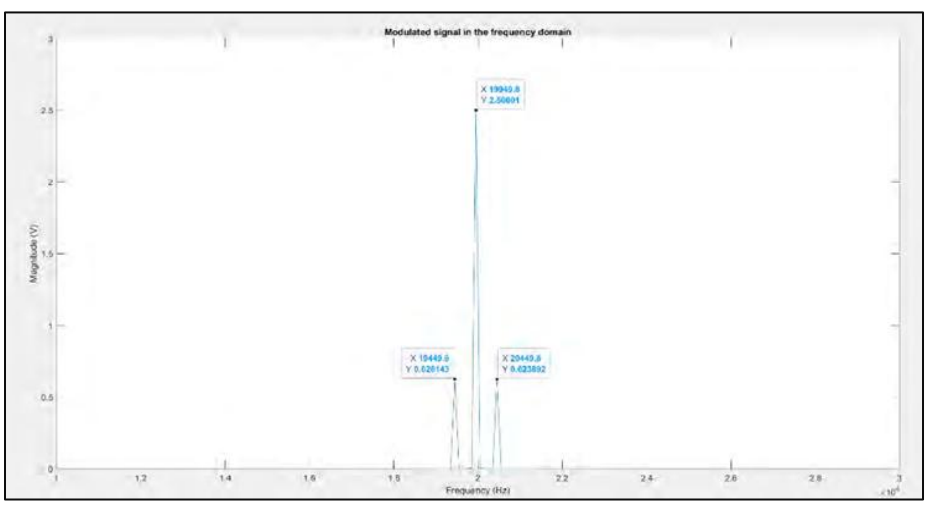

Figure 2. Modulated Signal in Frequency domain

b) Design a first and a third order low pass filter (butterworth filter) to demodulate the signal. The cut-off frequencies of the filters should be 1KHz. Plot the Bode diagram of these filters for a frequency range from 100Hz to 100KHz to verify the function.

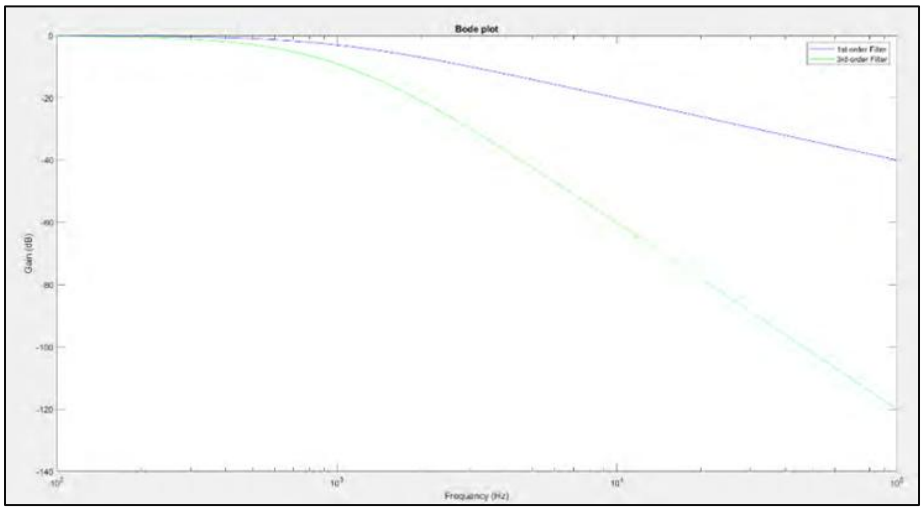

Figure 3. Bode Plot of 1st and 3rd order Filters

c) Rectify the AM modulated signal and apply the 1. order low pass filter to the rectified signal. Plot the rectified and the demodulated signal.

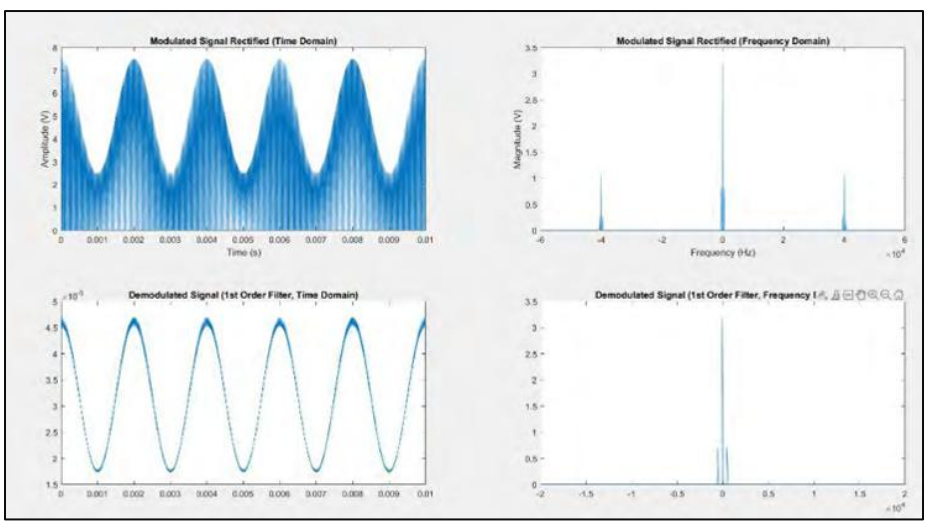

Figure 4. AM Modulated Signal rectified and demodulated signal 1st order

d) Change the order of the filter from 1. to 3. Plot the demodulated signal.

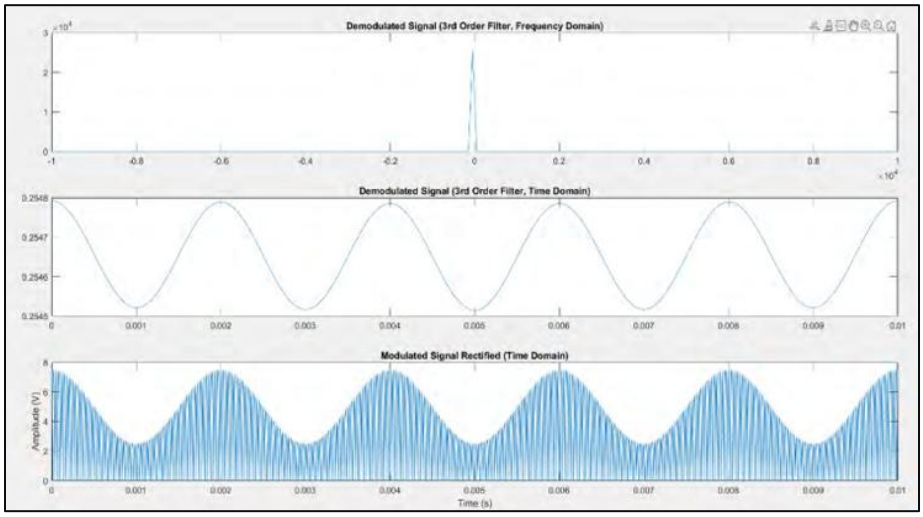

Figure 5. Demodulated Signal – 3rd order

e) Why is it better to use a higher order filter for the demodulation of the signal?

Using a higher order low-pass filter during the demodulation process provides a more accurate recovery of the original message signal. When an AM signal is rectified, several unwanted high-frequency components appear, particularly at twice the carrier frequency and its harmonics. A low-order filter, such as a first-order filter, only partially removes these components, resulting in residual high-frequency ripples in the demodulated signal.

In contrast, a higher order filter, such as a third-order filter, has a much steeper roll-off characteristic in the frequency domain, which allows it to attenuate unwanted frequencies far more effectively. This results in a smoother and cleaner output waveform that closely resembles the original message signal. Additionally, higher order filters contribute to better noise suppression and improved overall signal quality, making them more suitable for accurate and efficient demodulation of amplitude-modulated signals.

f) Attach the full Matlab script to the prelab.

```
% Parameters
fs = 10e6; % Sampling frequency
fc = 20e3; % Carrier frequency
fm = 500;  % Message frequency
A = 5;     % Carrier amplitude
m = 0.5;   % Modulation index
t = 0:1/fs:0.01; % Time vector (10ms)

% 1. Generate Signals
x = cos(2*pi*fm*t); % Modulating signal
c = A*cos(2*pi*fc*t); % Carrier signal
y = A* (1 + m * x) .* cos(2*pi*fc*t); % AM signal
```

```
% Plot Time Domain
figure('Name', 'Time Domain Signals');
subplot(3, 1, 1);
plot(t, x);
title('Modulating Signal');
xlabel('Time (s)'); ylabel('Amplitude (V)');
grid on;
subplot(3, 1, 2);
title('Carrier Signal');
xlabel('Time (s)'); ylabel('Amplitude (V)');
subplot(3, 1, 3);
plot(t, y);
title('Modulated Signal');
xlabel('Time (s)'); ylabel('Amplitude (V)');

% Plot Frequency Domain
figure('Name', 'Frequency Domain Signals');
L = length(y);
Y_fft = fftshift(fft(y) / L);
f = (-L/2:L/2-1) * (fs / L);
plot(f, abs(Y_fft));
xlim([1e4, 3e4]);
title('Modulated Signal in the Frequency Domain');
xlabel('Frequency (Hz)'); ylabel('Magnitude (V)');

% 2. Filter Design
figure('Name', 'Bode Diagram');
freq_range = 100:1:100e3;
w = 2*pi*freq_range;
R = 10e3 / (2*pi); % R and C for 1kHz cutoff
C = 100e-9;
H1 = 1 ./ (1 + 1i * w * R * C);
H3 = H1.^3;
semilogx(freq_range, 20*log10(abs(H1)), 'b');
hold on;
semilogx(freq_range, 20*log10(abs(H3)), 'r--');
legend('1st-order Filter', '3rd-order Filter');
title('Bode plot');
xlabel('Frequency (Hz)'); ylabel('Gain (dB)');

% 3. Rectify and Demodulate (1st Order)
figure('Name', 'Rectified and Demodulated Signals 1st Order');
rec = abs(y); % Rectify
subplot(2, 2, 1);
plot(t, rec);
title('Modulated Signal Rectified (Time Domain)');
xlabel('Time (s)'); ylabel('Amplitude (V)');
```

```
rec_spec = fftshift(fft(rec) / length(rec));
subplot(2, 2, 2);
plot(f, abs(rec_spec));
title('Modulated Signal Rectified (Frequency Domain)');
xlabel('Frequency (Hz)'); ylabel('Magnitude (V)');
xlim([-6e4, 6e4]);

% Apply 1st order filter in frequency domain
H1_full = 1 ./ (1 + 1i * 2*pi*f * R * C);
demod1_f = H1_full .* rec_spec;
demod1_t = ifft(ifftshift(demod1_f)) * length(rec);

subplot(2, 2, 4);
plot(f, abs(demod1_f));
xlim([-2e4, 2e4]);
title('Demodulated Signal (1st Order Filter, Freq Domain)');
subplot(2, 2, 3);
plot(t, abs(demod1_t));
title('Demodulated Signal (1st Order Filter, Time Domain)');

% 4. Demodulate (3rd Order)
figure('Name', 'Demodulated Signal 3rd Order Filter');
H3_full = H1_full.^3;
demod3_f = H3_full .* rec_spec;
demod3_t = ifft(ifftshift(demod3_f)) * length(rec);

subplot(3, 1, 1);
plot(f, abs(demod3_f));
xlim([-1e4, 1e4]);
title('Demodulated Signal (3rd Order Filter, Freq Domain)');
subplot(3, 1, 2);
plot(t, abs(demod3_t));
title('Demodulated Signal (3rd Order Filter, Time Domain)');
subplot(3, 1, 3);
plot(t, rec);
title('Modulated Signal Rectified (Time Domain)');
xlabel('Time (s)'); ylabel('Amplitude (V)');
```

Figure 6. MATLAB script for AM modulation and demodulation

2. Execution – Set-up and Results

Tools and Equipment

- 6n8F and 2 x 1µF Capacitors
- 22 kΩ and 180 kΩ resistors
- 1N4148 Diode
- LM741 Amplifier
- 10 mH Inductor
- Breadboard
- Oscilloscope for measurement
- Function Signal Generator

2.1. Execution Problem 1 – AM Modulated Signals in Time domain

The AM signal is generated by the function generator. Settings of the function generator:

Signal Shape – Sine
Modulation – AM
Carrier Frequency – 20 kHz
Carrier Amplitude – 10 V_{PP}
Modulation Frequency – 500 Hz
Modulation Index – 50%

The signal was directly connected to Channel 1 of the oscilloscope for measurement. The modulation index (m) on the signal generator was adjusted to three different values: 50%, 70%, and 120%. For each setting, the maximum and minimum envelope amplitudes (A_{max} and A_{min}) were measured using the oscilloscope's cursor function. To document the results, a hardcopy of each waveform was captured and saved for further analysis.

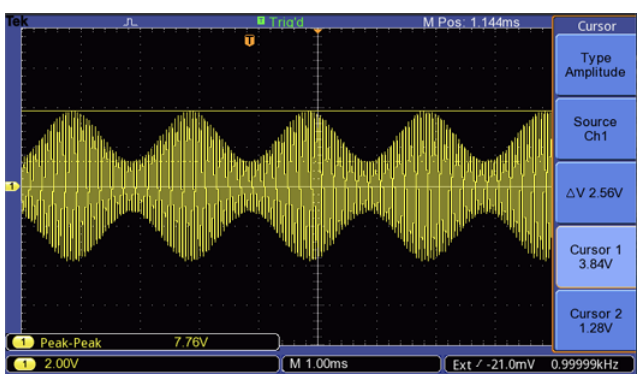

Figure 7. AM Modulated Sine Wave with 50% Modulation Index

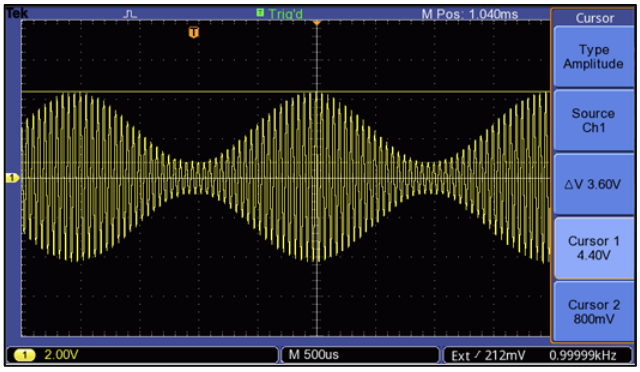

Figure 8. AM Modulated Sine Wave with 70% Modulation Index

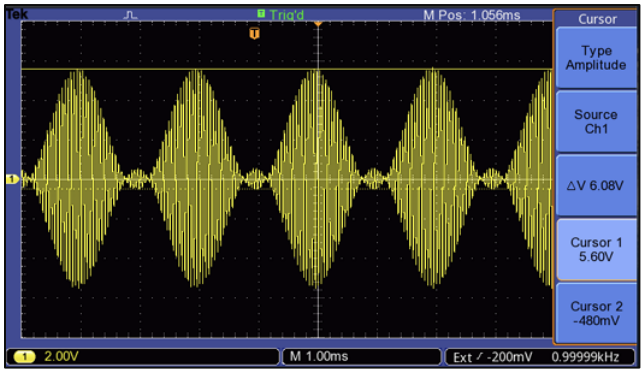

Figure 8. AM Modulated Sine Wave with 120% Modulation Index

Measuring the characteristics of an AM signal when the modulation index exceeds 100% is unreliable, as it leads to overmodulation. In this condition, the envelope of the waveform becomes distorted, making it impossible to accurately recover the original message signal during demodulation.

2.2. Execution Problem 2 – AM Modulated Signals in Frequency domain

The setup from the first part of the experiment was used again, with the modulation index adjusted to 70%. The oscilloscope's FFT function was utilized to display and analyze the frequency spectrum of the AM signal. Using the cursors, the frequency and magnitude of the carrier and its sidebands were measured. The obtained spectrum, illustrated in Figures 9 and 10, clearly shows three main components: a prominent carrier peak at approximately 19.81 kHz, a lower sideband at around 19.31 kHz, and an upper sideband at about 20.30 kHz. The frequency separation between the carrier and each sideband was measured to be roughly 500 Hz, which corresponds to the modulation frequency of the signal.

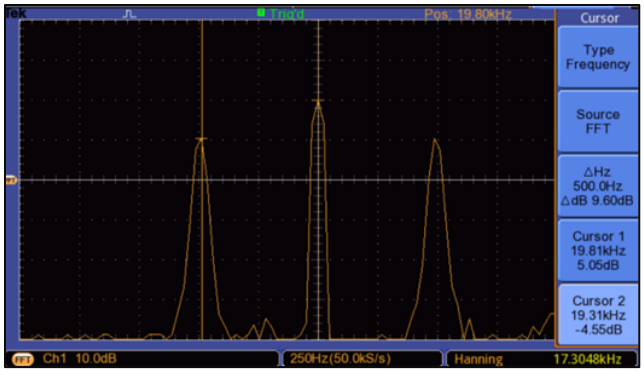

Figure 9. Magnitude and Frequency measurement of FFT of AM modulated Signal – Lower Sideband

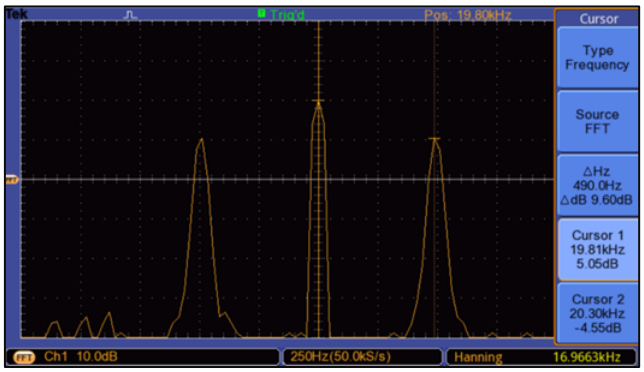

Figure 10. Magnitude and Frequency measurement of FFT of AM modulated Signal – Upper Sideband

2.3. Execution Problem 3 – Demodulation of a Message Signal

The demodulation circuit illustrated in Figure 11 was assembled for this part of the experiment. An LM741 operational amplifier was used and powered by a ±10V dual supply provided by the ELABO unit. To ensure circuit stability and minimize noise, 100nF decoupling capacitors were connected across the power supply lines, as shown in Figure 12.

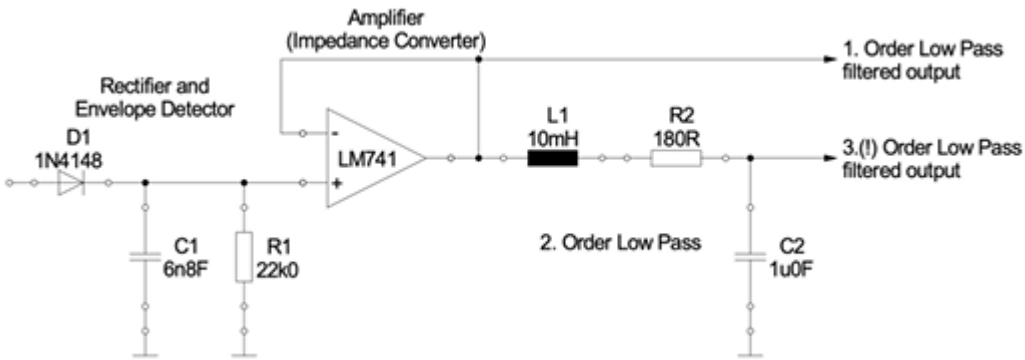

Figure 11. Demodulation Circuit

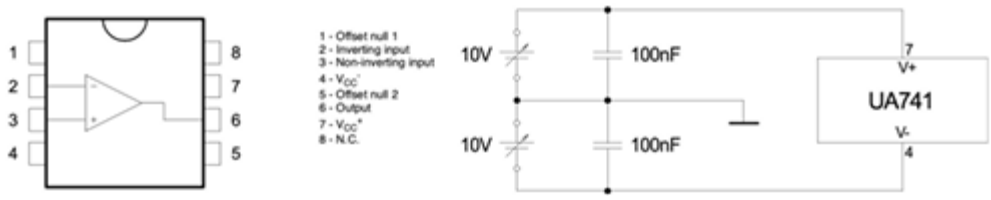

Figure 12. Op-Amp Supply Circuit

The Function Generator was connected to the input of the demodulating circuit and the following settings were used:

Signal Shape – Sine
Modulation – AM
Carrier Frequency – 20 kHz
Carrier Amplitude – 10 V_{PP}
Modulation Frequency – 500 Hz
Modulation Index – 50%

The Hardcopies of the AM modulated signal together with the 1st and 3rd order filter output was takes as the FFT spectrum of the 3rd order filter Output.

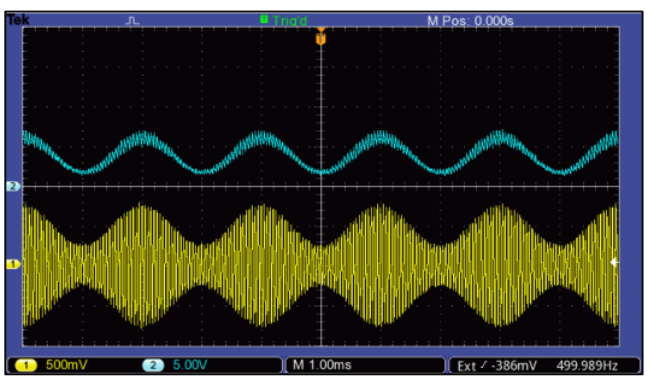

Figure 13. AM Modulated Signal and the 1st order filter output

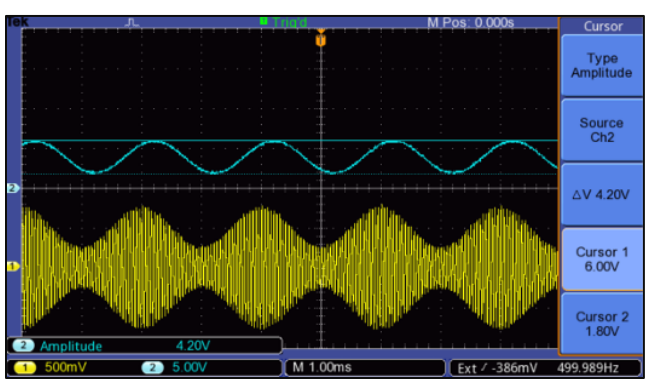

Figure 14. AM Modulated Signal and the 3rd order filter output

Finally, the FFT of the 3rd-order filter output was recorded, as shown in Figure 15. The resulting spectrum displayed a single prominent peak at approximately 495 Hz, corresponding to the original message frequency, with no visible component at 20 kHz. This confirms that the demodulation process was successful and that the higher-order filter efficiently eliminated the carrier component from the signal.

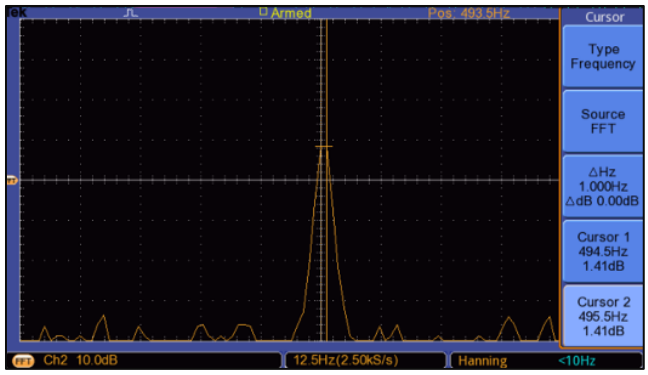

Figure 15. FFT of the 3rd order filter output, showing only the 500 Hz message signal

3. Evaluation

3.1. Evaluation Problem 1 – AM Modulated Signals in Time Domain

a) What is the relation between the modulation index and the relative magnitudes of the frequency components?

The relationship between the modulation index and the relative amplitudes of the frequency components can be determined using the following expression:

$$m = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

b) Calculated the modulation index using the measurements! Compare to the index you have used to generate the AM signal.

Values for A_{max} and A_{min} were taken from the oscilloscope plots using the cursor:

Generator 50%: $A_{max} = 3.84\ V, A_{min} = 1.28\ V$

$$m = \frac{3.84 - 1.28}{3.84 + 1.28} = 0.5 = 50\%$$

Generator 70%: $A_{max} = 4.40\ V, A_{min} = 0.8\ V$

$$m = \frac{4.40 - 0.8}{4.40 + 0.8} = 0.692 = 69.2\%$$

Generator 120%: $A_{max} = 5.60\ V, A_{min} = -0.48\ V$

$$m = \frac{5.60 - (-0.48)}{5.60 + (-0.48)} = 1.1875 = 118.8\%$$

c) Discuss the effect and the disadvantages of using a modulation index greater than 100.

When the modulation index exceeds 100%, the system enters a state known as overmodulation. In this condition, the envelope of the AM signal becomes distorted, as seen in Figure 8, where the lower part of the waveform clips and undergoes phase reversal. This distortion occurs because the instantaneous amplitude of the modulating signal exceeds that of the carrier, causing the envelope to lose its proper shape. As a result, an envelope detector can no longer correctly follow the signal's variations, producing a demodulated output that is heavily distorted and contains unwanted harmonic components. Therefore, to maintain accurate transmission and faithful demodulation, the modulation index must be kept below unity ($m \leq 1$).

3.2. Evaluation Problem 2 – AM Modulated Signals in Frequency Domain

a) How does the spectrum look like in theory? Compare to the experiment!

In theory, the frequency spectrum of a sinusoidally modulated AM signal consists of three discrete frequency components: a carrier at fc, and two sidebands located symmetrically at fc+fm and fc−fm . These appear as infinitely narrow spectral lines, representing ideal delta functions with no width. In the experimental spectrum, shown in Figures 9 and 10, the same three components are clearly visible at approximately 20 kHz (carrier), 19.5 kHz (lower sideband), and 20.5 kHz (upper sideband), confirming the theoretical prediction. However, the peaks observed experimentally are not perfectly sharp but slightly broadened. This broadening, or "spectral leakage," occurs due to the finite sampling window and the windowing function (such as Hanning) applied by the oscilloscope's FFT algorithm. Despite this, the experimental results closely align with the theoretical model, validating the expected frequency composition of the AM signal.

b) Does the function generator generate a DSB or DSB-SC AM signal?

The function generator produces a standard Double Sideband (DSB) AM signal, also referred to as DSB with Transmitted Carrier (DSB-TC). This is evident from the frequency spectrum shown in Figure 9, where a strong carrier component appears at the carrier frequency fcf_cfc, along with the two sidebands at fc+fm and fc−fm . If the signal were Double Sideband Suppressed Carrier (DSB-SC), the carrier peak would be absent, and only

the two sidebands would be visible. Therefore, the clear presence of a dominant carrier component confirms that the generator outputs a conventional DSB-TC AM signal.

c) How does changing the carrier frequency affect the AM spectrum?

Changing the carrier frequency fc affects the AM spectrum by shifting the entire set of frequency components along the frequency axis without altering their relative spacing. Specifically, both sidebands located at fc+fm and fc−fm move together with the carrier. For instance, if the carrier frequency is increased from 20 kHz to 30 kHz, the spectral peaks would appear at 29.5 kHz, 30 kHz, and 30.5 kHz. The overall shape of the spectrum remains unchanged, and only its position on the frequency axis is affected.

d) How does changing the message frequency affect the AM spectrum?

Changing the message frequency fm affects the AM spectrum by altering the distance between the carrier and its sidebands. The carrier component remains fixed at fc, while the sidebands shift to new positions at fc+fm and fc−fm. As the message frequency increases, the sidebands move further away from the carrier, widening the spectrum. Conversely, if fm decreases, the sidebands move closer to the carrier, resulting in a narrower spectral separation. For example, increasing fm from 500 Hz to 1 kHz shifts the sidebands from 19.5 kHz and 20.5 kHz to 19 kHz and 21 kHz, respectively.

e) Determine the modulation index m using the measured values.

The modulation index mmm can be determined using the measured amplitudes of the carrier and sideband components from the frequency spectrum. The relationship between these amplitudes is given by:

$$m = 2 * \frac{A_{sideband}}{A_{carrier}}$$

Since the oscilloscope displays the amplitudes in decibels (dB), the linear ratio can be expressed as:

$$\frac{A_{sideband}}{A_{carrier}} = 10^{(A_{sideband(dB)} - A_{sideband(dB)})/20}$$

Thus,

$$m = 2 * 10^{(A_{sideband(dB)} - A_{sideband(dB)})/20}$$

From the experimental data (Figure 9), the carrier amplitude is approximately 5.05 dB and the sideband amplitude is −4.55 dB. Substituting these values:

$$x = -4.55 - 5.05 = -9.6\ dB$$

$$m = 2 * 10^{-\frac{9.6}{20}} = 2 * 10^{-0.48} = 0.662$$

$$m = 66.2\ \%$$

This calculated modulation index closely matches the 70% modulation setting on the signal generator, with the small deviation likely due to measurement limitations and instrument precision.

3.3 Evaluation Problem 3 – Demodulation of a message signal

a) Compare the 1st and 3rd order filter output signal with the message signal.

When comparing the demodulated outputs obtained using the 1st-order and 3rd-order filters with the original message signal, noticeable differences can be observed in both noise level and phase behavior. The 1st-order filter produces a demodulated signal that still contains a considerable amount of high-frequency noise, resulting in a waveform that only roughly follows the shape of the original message signal. However, its phase distortion is relatively small. In contrast, the 3rd-order filter provides a much cleaner output with significantly reduced noise, yielding a waveform that more closely matches the original message in both shape and amplitude. The improvement in smoothness comes at the cost of a slightly increased phase shift, which causes a small delay between the demodulated and the original message signals. Overall, the 3rd-order filter provides superior fidelity in amplitude recovery, while the 1st-order filter is simpler but less effective at suppressing high-frequency components.

b) Compare the measured signals with the MatLab results. What are the differences between simulation and measurement?

The measured signals and MATLAB results show similar overall behavior, confirming the experiment's validity. However, the measured signals contain noticeable noise and slight phase shifts, while the MATLAB outputs are clean and perfectly aligned. These differences arise from real-world factors such as component tolerances, circuit parasitics, and the non-ideal behavior of the op-amp and instruments. MATLAB provides ideal, noise-free results, whereas the experimental setup introduces small distortions and delays typical of practical circuits.

4. Conclusion

This experiment successfully demonstrated the fundamental principles of Amplitude Modulation (AM), covering signal generation, frequency analysis, and demodulation. In the first part, the time-domain behavior of AM was investigated by varying the modulation index to 50%, 70%, and 120%. The results confirmed that increasing the modulation index deepens the modulation effect, while overmodulation (m > 100%) causes clear envelope distortion, making the signal unreliable.

In the second part, the AM signal was analyzed in the frequency domain using the oscilloscope's FFT function. The measured spectrum showed the expected three main components the carrier and its upper and lower sidebands consistent with theoretical predictions. Minor deviations in frequency and amplitude were attributed to measurement limitations and equipment accuracy.

The final part of the experiment focused on demodulation of the AM signal using both 1st- and 3rd-order low-pass filters. The 3rd-order filter provided a smoother and cleaner output by more effectively suppressing the carrier and high-frequency components, whereas the 1st-order filter left noticeable noise in the output. Comparison with MATLAB simulations confirmed the experimental trends, though the simulated signals appeared ideal due to the absence of non-ideal circuit effects such as noise, component tolerances, and phase shifts.

Overall, the experiment provided a complete and practical understanding of AM modulation and demodulation. It highlighted how theoretical principles translate into real-world circuits and demonstrated the influence of modulation depth, filtering order, and system non-idealities on signal quality.

5. References

1. 20240829-co-520-b manual- U.Pagel, Page(65-84) http://uwp-cu-lab.my-board.org/02.0.adveelab/02.1.signalsys/20240829-co-520-b manual.pdf

6. Appendix

6.1. Prelab

6.1.1. Prelab Problem 1 – Frequency Modulation

A sinusoidal modulation signal,

$$m(t) = 4\cos(8000\pi t) \qquad K_f = 10\ kHz/V$$

Frequency Deviation: $\Delta f = K_f A_m = 10 * 4 = 40\ kHz$

Frequency of the modulating signal: $f_m = \frac{8000\pi}{2\pi} = 4\ kHz$

Frequency modulation index: $\beta_f = \frac{K_f A_m}{f_m} = \frac{40}{4} = 10$

6.1.2. Prelab Problem 2 – FM Signal in the frequency domain

An FM signal has A_C =2.5 V, f_C = 40 kHz and f_m=5 kHz. We plot the spectrum for $\beta_f = 0.2, 1\ and\ 2$.

Carson's Rule: $B_T = 2f_m(\beta_f + 1)$

$$for\ \beta_f = 0.2: B_T = 2 * 5 * (0.2 + 1) = 12\ kHz$$

$$for\ \beta_f = 1: B_T = 2 * 5 * (1 + 1) = 20\ kHz$$

$$for\ \beta_f = 2: B_T = 2 * 5 * (2 + 1) = 30\ kHz$$

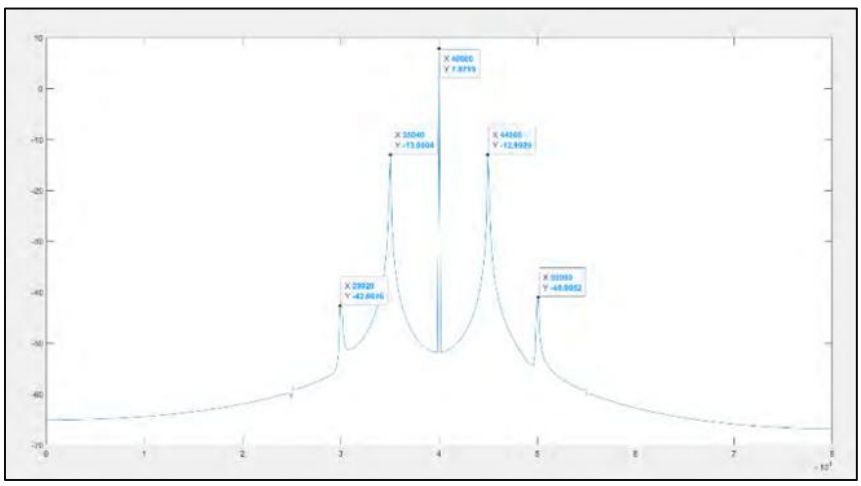

Figure 16. FM Spectrum for $\beta_f = 0.2$

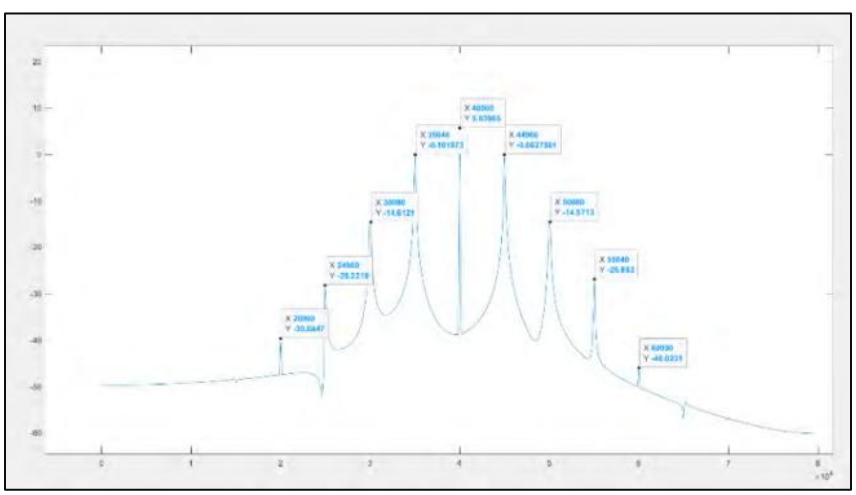

Figure 17. FM Spectrum for $\beta_f = 1$

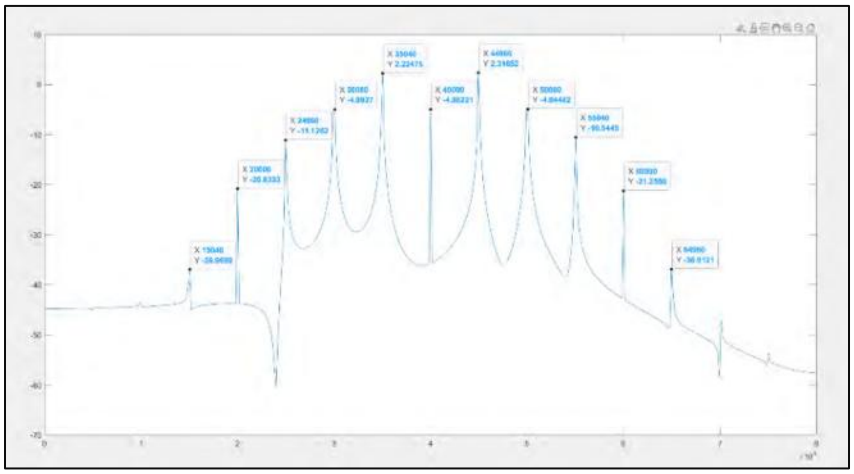

Figure 18. FM Spectrum for $\beta_f = 2$

	$15kHz$	$20kHz$	$25kHz$	$30kHz$	$35kHz$	$40kHz$	$45kHz$	$50kHz$	$55kHz$	$60kHz$	$65kHz$
$\beta_f = 0.2$	-	-	-	-40.98	-13.01	7.87	12.99	-40.99	-	-	-
$\beta_f = 1$	-	-39.68	-28.22	-14.88	-0.10	5.64	-0.06	-14.57	-26.88	-46.02	-
$\beta_f = 2$	-36.97	-20.83	-11.16	-5.02	2.22	-4.98	2.31	-4.94	-10.54	-21.26	-36.91

Table 1. Peak magnitude inside the bandwidth from all plots

6.1.3. Prelab Problem 3 – Frequency Demodulation

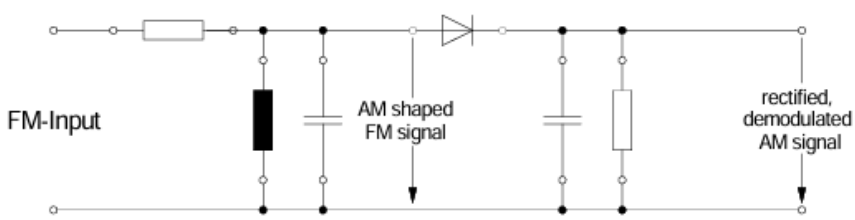

Figure 19. Slope detector Circuit

The slope detector is a simple type of FM demodulator that works by converting a frequency-modulated signal into an amplitude-modulated signal, which can then be demodulated using a standard AM envelope detector.

The circuit operates in two main stages:

1. FM-to-AM Conversion:

   The first stage uses a resonant RLC or LC circuit that is deliberately detuned so that its

resonant frequency is slightly offset from the FM carrier frequency. As the instantaneous frequency of the FM signal varies around the carrier, it moves along the slope of the filter's frequency response curve. Because the circuit's output amplitude depends on the input frequency, these small frequency variations are converted into amplitude variations. This effectively produces an AM-shaped signal whose amplitude follows the original message signal.

2. AM Demodulation:

   The second stage is a diode-based envelope detector consisting of a diode and an RC low-pass filter. This stage rectifies the AM-shaped signal and removes the high-frequency carrier, leaving only the low-frequency envelope, which corresponds to the original message signal.

In summary, the slope detector demodulates an FM signal by converting frequency variations into amplitude variations using a tuned circuit, and then recovering the message signal with an envelope detector.

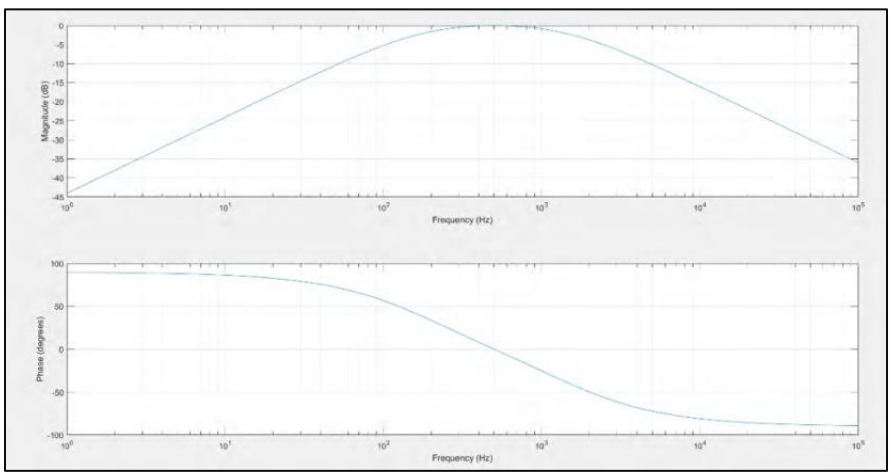

Figure 20. Bode diagram of the RLC Slope Detector, magnitude and phase